1. Product Overview

Product name: StudyAI
Type: Web-based AI reading comprehension platform for students and educators.
Product owner: Product team
Status: Draft for pitching / v1.0 concept

Elevator pitch
StudyAI transforms any text (stories, articles, textbook chapters, PDFs, slides) into adaptive, exam-style comprehension quizzes with instant feedback, analytics, and classroom features. It helps students learn more actively and helps teachers save time on assessment creation.

Vision
Become the default “assessment layer” on top of digital reading: wherever a student reads, StudyAI can generate high-quality questions and insights in seconds.

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2. Goals and Success Metrics

Business goals (first 12–18 months)
- Reach meaningful adoption in schools and self-study users.
- Prove that adaptive comprehension practice improves reading scores and engagement.
- Build a platform that can later expand into vocabulary, grammar, and content recommendations.

Key success metrics
- Activation:
  - ≥ 70% of sign-ups generate at least 1 quiz in their first session.
- Engagement:
  - Median user generates ≥ 4 quizzes per week (students) or ≥ 10 per month (teachers).
- Retention:
  - Weekly retention ≥ 40% for student accounts.
  - Monthly retention ≥ 50% for teachers.
- Learning impact (later phase):
  - Users who complete ≥ 20 quizzes show measurable improvement on standardized-style comprehension questions compared to baseline.

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3. Target Users and Personas

Primary personas
- **Student Sam (grades 6–12):**

  - Uses StudyAI to practice for school tests and standardized exams.
  - Wants fast, accurate feedback, and some gamification (XP, streaks, leaderboards).
- **Teacher Taylor (grades 4–12):**
  - Needs to generate quizzes for class readings, homework, and quick checks for understanding.
  - Wants control over difficulty, question types, and export formats (Google Forms, LMS, print).
- **Tutor Tina (private tutor / test prep):**
  - Uses StudyAI to generate tailored practice sets for individual students.
  - Wants analytics across multiple students and over time.

Secondary: Edtech companies that might embed StudyAI via API.

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4. Problem and Strategic Fit

Problems
- Writing high-quality comprehension questions is time-consuming and uneven in quality.
- Students rarely get enough varied, exam-like practice tied directly to what they read.
- Existing AI quiz generators are generic, lack pedagogy, or focus on flashcards rather than reading comprehension.

Strategic fit
- Leverages modern LLMs for question generation, scoring, and explanations.
- Sits at the intersection of AI, K–12/early college education, and productivity tools for teachers.

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5. Core Value Proposition

1. **From any text to quiz in seconds**: Paste text or upload files, click generate, get a structured quiz.
2. **Pedagogically sound questions**: Mix of literal, inferential, main idea, author's purpose, and vocabulary-in-context.
3. **Adaptive difficulty and analytics**: The system learns which skills a student struggles with and targets them.
4. **Teacher & classroom workflows**: Easy sharing, exports, and tracking of student performance.

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6. Scope and Feature Set (v1.0)

6.1 Input and Content Handling

- Accept input via:
  - Pasted text.
  - File upload: PDF, DOCX, PPTX (basic text extraction).
  - URL (for accessible articles/blog posts).
- Text pre-processing:
  - Detect language and reading level.
  - Clean formatting, remove headers/footers, etc.

6.2 Quiz Generation

- Question types:
  - Multiple choice (1 correct, 3–4 distractors).
  - Short answer (1–3 sentences).
  - “Evidence-based” questions (user must select the supporting sentence/paragraph).
- Cognitive categories:
  - Detail, main idea, inference, author’s purpose, tone, vocabulary-in-context, structure/organization.
- Difficulty control:
  - Level (elementary, middle, high school, test-prep).
  - The system adapts question complexity and vocabulary.
- User controls:
  - Number of questions (5–25).
  - Mix of question types and categories (e.g., “emphasize inference and main idea”).

6.3 Quiz Taking Experience

- Clean, distraction-free quiz UI optimized for web and tablet.
- Modes:
  - Practice mode: immediate feedback after each question.
  - Test mode: feedback only at the end, with timer and optional time limit.
- Accessibility:
  - Keyboard navigation, screen-reader-friendly markup, adjustable font size.

6.4 Scoring, Explanations, and Feedback

- Automatic scoring for MCQs; partial credit logic for multi-part questions.
- Short-answer scoring: semantic similarity and rubric-based grading, with teacher override.
- For each question:
  - Correct answer.
  - Short explanation and, for some, “where in the text” the evidence is.
- Skill breakdown per quiz:
  - Example: “Main idea: 3/5, Inference: 1/4, Vocabulary: 4/4.”

6.5 Analytics and Progress

- Student dashboard:
  - Recent quizzes, average score, time spent, skill heatmap.
  - Streaks and simple gamification (XP, levels, badges).
- Teacher dashboard:
  - Classes/rosters.
  - Per-student performance and per-skill performance over time.
  - Exportable reports (CSV, PDF) for gradebook/LMS.

6.6 Sharing, Export, and Integrations

- Quiz export options:
  - Share link (students complete quiz in StudyAI).
  - Export to Google Forms.
  - Export to PDF (printable).
- Authentication and roles:
  - Student, Teacher, Admin roles.
- Later phase (v1.x): LMS integrations (e.g., Google Classroom, Canvas) via LTI or APIs.

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7. User Stories (Samples)

- As a **student**, I want to generate quizzes from my assigned reading so that I can prepare for tests more effectively.
- As a **teacher**, I want to control question difficulty and type so that quizzes match my students' level.
- As a **teacher**, I want to see which skills my class struggles with so that I can adjust my instruction.
- As a **tutor**, I want to assign quizzes to individual students and track their improvement over time.

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8. Non-Functional Requirements

- **Performance:**
  - Generation under ~10 seconds for typical passages; show progressive loading.
- **Security & privacy:**
  - FERPA/GDPR-friendly data handling where possible; encrypted in transit and at rest.
- **Scalability:**
  - Architecture that can support classroom/ school-level usage (hundreds of concurrent quiz-takers).

- **Reliability:**
  - Graceful degradation if AI model is slow/unavailable (e.g., “retry” and clear messaging).

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9. Competition and Differentiation

Examples of related tools include AI quiz generators that turn notes or PDFs into quizzes, and live quiz platforms that focus on in-class polling and leaderboards.

Differentiation:
- Deep focus on reading comprehension, not generic Q&A.
- Skill-based analytics and adaptive generation.
- Strong teacher workflows and exports, not just a “solo AI toy.”

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10. Out of Scope (for now)

- Full-blown curriculum or content library (StudyAI focuses on user-provided text).
- Native mobile apps (web first, responsive for tablets).
- Speech-based reading or audio comprehension.

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11. Open Questions / Risks

- How strict should the product be about age verification / data privacy for younger students?
- How much control should teachers have over editing AI-generated questions before assigning?
- How to prove learning impact in a way that’s convincing to schools (A/B tests, pilots)?